

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 10 • Radians

Questions

Student book • questions with writing space

- 64 pages
- Past papers up to June 2026

Radians

Your classified route

19
questions

173
marks

4
families

12
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Radians

Families and variants

F01 • Sector measures and inverse sector problems

V01 Direct arc, area and perimeter of one sector

V02 Recover radius or angle from arc length or sector area

V03 Concentric, annular and linked sectors

V04 Coupled Sector Area-and-Perimeter Constraints

F02 • Recovering the central angle and choosing the correct arc

V01 Angle from straight lines, symmetry or rotational partition

V02 Angle from triangle trigonometry

V03 Major, minor and reflex sector selection

F03 • Composite sector designs

V01 Minor sector joined to one or more polygons

V02 Major sector with an attached or removed polygon

V03 Repeated rotational sector design

F04 • Circular segments and chords

V01 Segment area

V02 Segment perimeter and chord

Sector measures and inverse sector problems

WHAT THIS FAMILY ASKS

Apply $s = r \theta$ and $A = \frac{1}{2} r^2 \theta$ directly or inversely, including linked and concentric sectors.

VARIANT ROUTE

V02 Recover radius or angle from arc length or sector area

V03 Concentric, annular and linked sectors

V04 Coupled Sector Area-and-Perimeter Constraints

5 Complete Classified questions

37 total marks

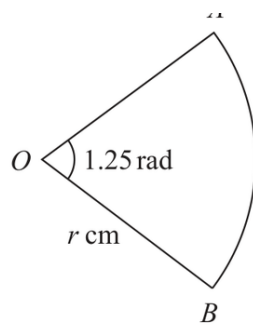


Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

- (a) find the exact value of r ,

(2)



The angle AOB is 1.25 radians.

(b) find the exact length of the perimeter of the sector. Write your answer in simplest form.

(3)



Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(1)

Diagram **NOT**
accurately drawn

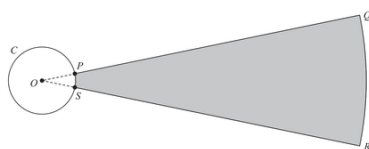


Figure 3

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

$$5x^2 + 12x - 1500 = 0$$

(3)



The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA:AO = 3:5$

(3)

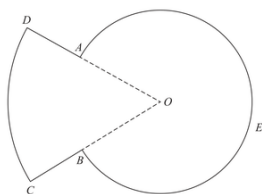


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector AOB of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA:AO = 3:5$

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

DR ESAM AHMED

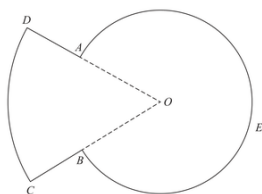


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector AOB of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA:AO = 3:5$

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

DR ESAM AHMED



Figure 2 shows the plan view of a design for a swimming pool.

Given that

- (a) show that, to 3 significant figures, $OQ = 8.83 \text{ m}$

(3)

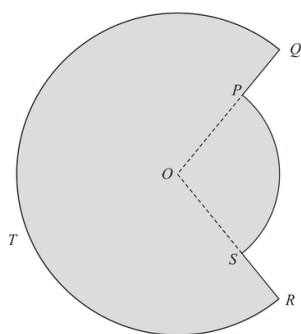


Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

$$OQ = 8.83 \text{ m}$$

(b) Find the total surface area of the swimming pool in m^2 to the nearest integer.

(3)

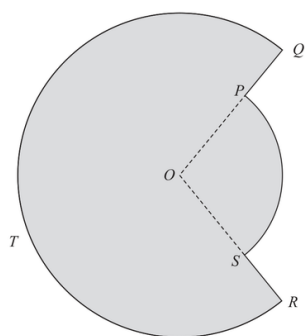


Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

$$OQ = 8.83 \text{ m}$$

(c) Find the total perimeter of the swimming pool in metres to 2 significant figures.

(3)

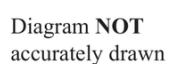


Figure 3 shows a sector AOB of a circle, centre O .

The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

(a) write down two different equations involving r and θ .

(2)

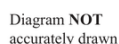


Figure 3 shows a sector AOB of a circle, centre O .

The radius OA of the sector is r cm and angle AOB is θ radians.

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

(b) solve the two simultaneous equations in part (a) to find the exact value of r and the exact value of θ .

Write each answer in the form $a + b\sqrt{c}$ where a , b and c are integers.

(Solutions relying on calculator technology are not acceptable.)

(5)

Recovering the central angle and choosing the correct arc

WHAT THIS FAMILY ASKS

Obtain the radian angle from a diagram before applying sector formulae, and decide whether the required region uses θ or 2π minus θ .

VARIANT ROUTE

V02 Angle from triangle trigonometry

2 Complete Classified questions

19 total marks

Figure 3 shows the plan view of a viewing platform at a tourist site.

Given that

- find

(3)

Figure 3 shows the plan view of a viewing platform at a tourist site.

Given that

- $OA = OC = 6\text{ m}$
- $AD = 14\text{ m}$
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

(b) the length of arc ABC , in metres, to one decimal place,

Figure 3 shows the plan view of a viewing platform at a tourist site.

Given that

- $OA = OC = 6\text{ m}$
- $AD = 14\text{ m}$
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

(c) the total area of the viewing platform, in m^2 , to one decimal place.

(4)

The diagram shows a circle with center O . A sector OAC is defined by radii OA and OC and a central angle θ . A point B is located to the right of the circle. Line segments AB and BC are drawn, connecting the points A and C on the circle to B . A line segment OB is also drawn, passing through the center O . The region X is the area bounded by the arc AC and the line segments AB and BC .

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

- $OA = OC = 0.6\text{ m}$
- $AB = CB = 1.4\text{ m}$
- Angle $OAB = \text{Angle } OCB = 2\text{ radians}$
- Angle $AOC = \theta\text{ radians}$, as shown in Figure 3

(a) show that $\theta = 1.64$ radians to 3 significant figures,

(4)

7.

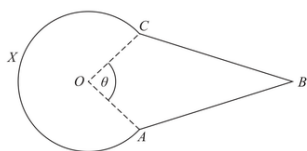


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector OCA of a circle centre O .

In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

(b) find the perimeter of the sign, in metres to 2 significant figures,

(2)

DR ESLAM AHMED

7.

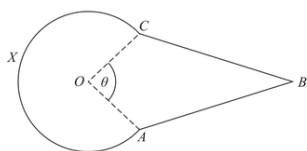


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector OCA of a circle centre O .

In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

(c) find the area of the sign, in m^2 to 2 significant figures.

(4)

DR ESLAM AHMED

Composite sector designs

WHAT THIS FAMILY ASKS

Assemble or subtract sector, triangle, trapezium, rectangle and radial-edge contributions to find a complete area or perimeter.

VARIANT ROUTE

V01 Minor sector joined to one or more polygons

V02 Major sector with an attached or removed polygon

V03 Repeated rotational sector design

11 Complete Classified questions

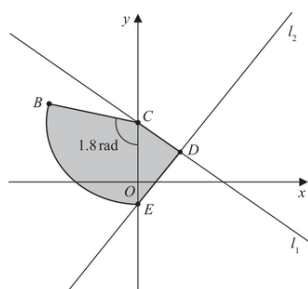
110 total marks



The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

- (a) State the y coordinate of C .

(1)



The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

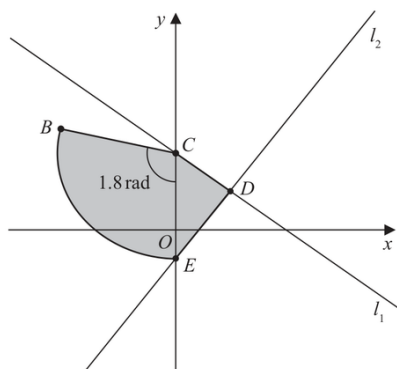
The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)



The line l_1 has equation $4y + 3x = 48$

The point $D(8, 6)$ lies on l_1

the y coordinate of E is $-\frac{14}{3}$

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

(d) Calculate the exact area of the region $CBED$.

(3)



The design consists of a sector OBC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(2)



The design consists of a sector OBC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(1)



The design consists of a sector OBC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(c) the total area of the stage, giving your answer, in m^2 , to one decimal place,



The design consists of a sector OBC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(d) the total perimeter of the stage, giving your answer, in m, to one decimal place.

(4)



In the plan

- (a) Show that angle AOB is 1.186 radians to 4 significant figures.

(2)



In the plan

- (b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.



In the plan

- (c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

Diagram NOT
necessarily drawn

The diagram shows a sector ABC with center B . The radius AB is 10.8 m and the angle ABC is 1.3 rad . A point E is on the arc AC such that $BE = 6\text{ m}$. A point D is on the arc AC such that $BD = 6\text{ m}$ and DC is perpendicular to BC . The length of the chord ED is 12.2 m .

Figure 2 shows the plan view of a garden.

The points A , B and C lie on a straight line with $AB = 10.8$ m

Angle $BCD = \frac{\pi}{2}$ radians, angle $EBD = 1.3$ radians and $AE = 12.2$ m

- (a) Find the area of the sector BDE , giving your answer in m^2

(2)

Diagram NOT
necessarily drawn

Diagram of a bridge cross-section ABC . The base is a horizontal line segment AC . Point B is on AC . A solid line segment AB has a length of 10.8 m . A solid line segment BE has a length of 12.2 m . A dashed line segment BL has a length of 6 m . A dashed line segment EL has a length of 6 m . The angle EBL is 1.3 rad . A vertical line segment LC is drawn from L to C , with a right-angle symbol at C . A curved line segment connects E and L , representing the bridge arch.

Figure 2 shows the plan view of a garden.

The points A , B and C lie on a straight line with $AB = 10.8$ m

(b) Find the size of angle ABE , giving your answer in radians to 2 decimal places.

(2)

Diagram NOT
necessarily drawn

Diagram illustrating a bridge cross-section. The bridge is supported by a horizontal base line AC . A point B lies on AC . The distance AB is 10.8 m . A solid line segment BE represents the bridge structure, with a length of 12.2 m . A dashed line segment BE is drawn, with a length of 6 m . A dashed line segment EL is drawn, with a length of 6 m . A dashed line segment BL is drawn, with a length of 6 m . The angle EBL is 1.3 rad . A vertical line segment LC is drawn, perpendicular to AC at C , indicated by a right-angle symbol.

Figure 2 shows the plan view of a garden.

The points A , B and C lie on a straight line with $AB = 10.8$ m

(c) Find the area of the garden, giving your answer in m^2 to 3 significant figures.

(3)

5. A plot of land OAB is in the shape of a sector of a circle with centre O .

Given

- $OA = OB = 5 \text{ km}$
- angle $AOB = 1.2 \text{ radians}$

(a) find the perimeter of the plot of land.

(2)

DR ESLAM AHMED

The diagram shows a circular sector OAB with center O and radius 5 km . The central angle $\angle AOB$ is 1.2 rad . A point P is located on the radius OB , and a line segment AP is drawn. The region R_1 is the circular sector APB , and the region R_2 is the triangle OAP . The area of R_1 is given as 1.5 km^2 .

Figure 2

Given that

(b) show that the area of $R_2 = 3.75 \text{ km}^2$

(3)

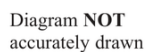


Figure 2

area of $R_2 = 3.75 \text{ km}^2$

(4)

The diagram shows a sector \$OACB\$ with vertex \$O\$. The radius \$OC\$ is \$3.4\$ cm. The angle \$AOB\$ is \$\frac{\pi}{6}\$. A point \$B\$ is located on the arc \$AC\$ such that the line segment \$OB\$ has a length of \$1.9\$ cm. The region bounded by the arc \$AC\$ and the line segments \$OA\$, \$AB\$, and \$BC\$ is shaded.

Figure 1 shows a sketch of a design for a badge.
The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- Making your method clear,

(3)



- $OB = 3.4$ cm
- $AB = 1.9$ cm
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(2)

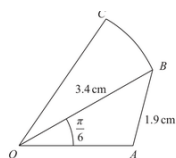


Figure 1

Figure 1 shows a sketch of a design for a badge.
The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- $OB = 3.4\text{ cm}$
- $AB = 1.9\text{ cm}$
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

Given that the ratio of the area of sector OBC to the area of triangle OAB is $3:2$

(c) show that angle BOC is 0.462 radians to 3 significant figures.

(3)

DR ESAM AHMED



The design consists of a triangle OAB joined to a sector OBC of a circle with centre O . In the design

(d) Hence find the perimeter of the badge, in cm, to the nearest integer.

(5)

Not to scale

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$APCOA$ is a sector of a circle centre O with radius 12 m.

(a) Show that the size of angle AOD is 1.231 radians to 3 decimal places.

(2)

Figure 1 shows a shaded sector of a circle with center O and radius 10 cm . The sector is formed by radii OA and OC , and the arc AC . The angle AOC is 90° . A rectangle is constructed with vertices A , B , C , and D , where D is on the radius OC . The rectangle is shaded gray. The text "Not to scale" is written next to the rectangle.

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$APCOA$ is a sector of a circle centre O with radius 12 m.

The lawn $APCDA$ is shown shaded in Figure 2.

(4)

Figure 1 shows a shaded sector of a circle with center O . The sector is bounded by radii OA and OC , and the arc AC . A dashed rectangle is constructed with vertices A , B , C , and D . Point D is located on the radius OC . The text "Not to scale" is present in the upper right corner of the diagram.

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$APCOA$ is a sector of a circle centre O with radius 12 m.

(c) Find the perimeter of the lawn, in metres, to one decimal place.

(3)

Diagram **NOT**
to scale

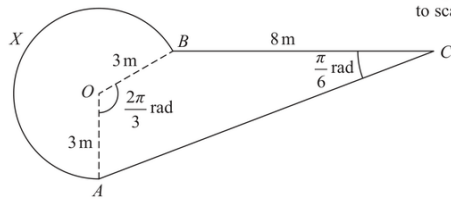


Figure 2 shows the plan view of a design for a pond.

- $BC = 8\text{ m}$

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,
(ii) the exact perimeter of the sector $AOBX$.

(5)

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- (b) Calculate the exact area of the triangle AOB .

(c) Show that the length AB is $3\sqrt{3}$ m.



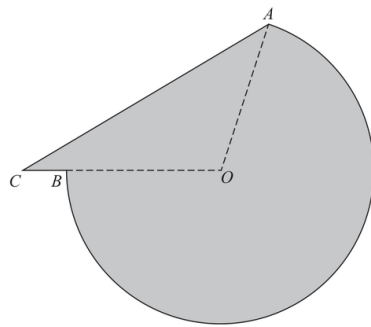
The shaded area in Figure 2 shows the plan view of a helicopter landing pad. The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

Given that

- $AO = OB = 15\text{ m}$
- $BC = 2\text{ m}$
- CBO is a straight line
- angle $ACO = 0.6$ radians

(a) show that angle COA is 1.847 radians to 3 decimal places.

(3)



The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

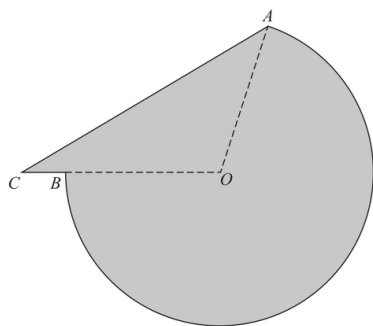
Given that

- $AO = OB = 15\text{ m}$
- $BC = 2\text{ m}$
- CBO is a straight line
- angle $ACO = 0.6$ radians

angle COA is 1.847 radians to 3 decimal places.

- (b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures.

(3)



The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

Given that

- $AO = OB = 15$ m
- $BC = 2$ m
- CBO is a straight line
- angle $ACO = 0.6$ radians

angle COA is 1.847 radians to 3 decimal places.

- (c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures.

(3)

5.

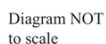


Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector $BODCB$ of a circle centre O joined to a triangle AOD .

Given that

- ABO is a straight line
- $OD = 3$ cm
- $AD = 8$ cm
- angle $AOD = 2.5$ radians

(a) show that angle ODA is 0.415 radians to 3 significant figures.

(3)

5.

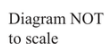


Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector $BODCB$ of a circle centre O joined to a triangle AOD .

Given that

- ABO is a straight line
- $OD = 3$ cm
- $AD = 8$ cm
- angle $AOD = 2.5$ radians

(b) Find the total area of the logo, in cm^2 , to one decimal place.

(3)

5.

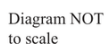


Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector $BODCB$ of a circle centre O joined to a triangle AOD .

Given that

- ABO is a straight line
- $OD = 3$ cm
- $AD = 8$ cm
- angle $AOD = 2.5$ radians

(c) Find the perimeter of the logo, $ABCD$, in cm, to one decimal place.

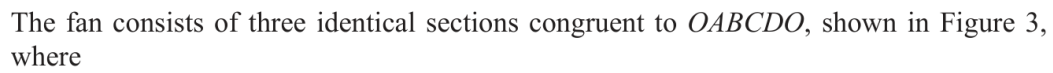
(3)

A diagram of a circular sector with center O . The sector is bounded by two radii and an arc. Points A , B , C , and D are marked on the arc. The radii are labeled OA and OB . The arc is labeled ACD .

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- $\text{angle } AOD = \frac{2\pi}{3}$ radians

(a) show that the length of the arc CD is 35.9 cm to one decimal place.



- Given that the length of the arc AB is 15 cm,
arc CD is 35.9 cm to one decimal place.

Find, according to the model,

(2)

DR ESLAM AHMED • ELITE MATHEMATICS

Circular segments and chords

WHAT THIS FAMILY ASKS

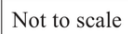
Use sector-minus-triangle area and arc-plus-chord perimeter for a circular segment.

VARIANT ROUTE

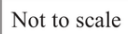
V01 Segment area

1 Complete Classified questions

7 total marks



(1)



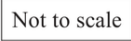


Figure 2 shows a plan view of a semicircular garden $ABCDEOA$

- centre O
- diameter AOE
- radius 3 m

(c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

(3)